



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2014

BIOLOGY

Higher 2

Paper 2 Core Paper

9648/02

Thursday, 28 August

2 hours

Candidate Name: _____

Civics Group: _____

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all answer papers used.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Part I and Part II)

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten the answer papers securely and hand up separately.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
Section A	
1	/12
2	/11
3	/11
4	/10
5	/10
6	/10
7	/ 9
8	/ 7
Section B	
9 / 10	/20
TOTAL	/100

This document consists of **23** printed pages and **1** blank page.

[Turn over

Section A [Part 1]For
Examiner's
UseAnswer **all** the questions in this section.

- 1** Fig. 1.1 shows an electron micrograph of part of a liver cell.

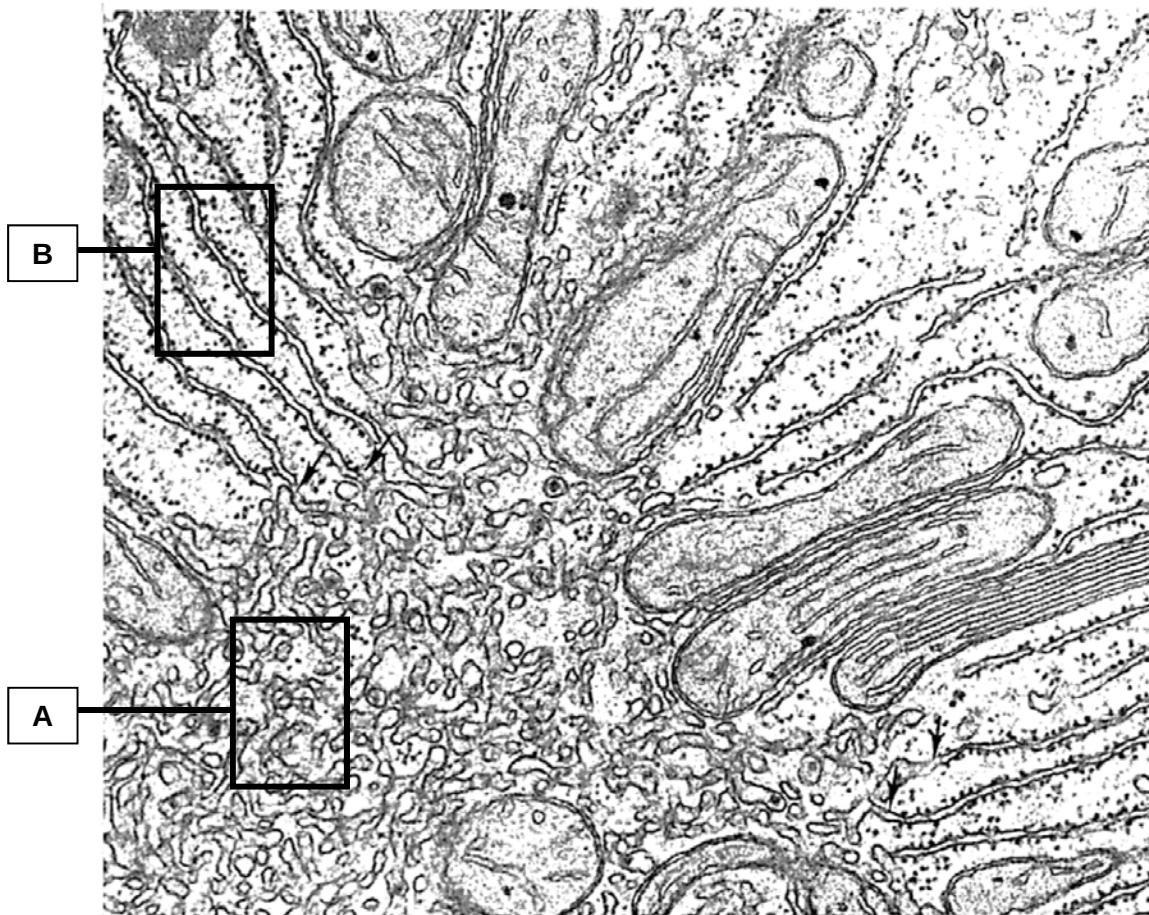


Fig. 1.1

- (a)** Identify structures **A** and **B**.

A: _____

B: _____

[2]

- (b)** State the function of structure **A**.

_____ [1]

[Turn over

(c) Relate the structural features of **B** to its function.

[2]

Cholesterol is taken up by liver cells, which transforms it into 30 to 100 nm low-density lipoprotein (LDL) particles. Fig. 1.2 shows the structure of an LDL particle. It consists of a core of triglycerides and a surface coat of phospholipids, cholesterol and a monomeric protein, ApoB-100.

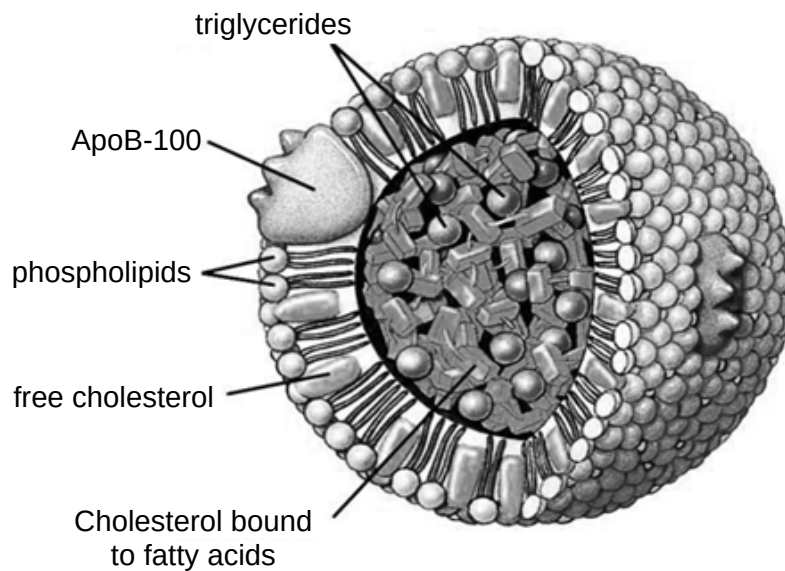


Fig. 1.2

(d) With reference to Fig. 1.2, outline the roles of lipids in the LDL coat.

[3]

(e) Fig. 1.3 shows the three dimensional structure of ApoB-100.

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Fig. 1.3

- (i) Label the type(s) of secondary protein structure observed in Fig. 1.3. [1]
- (ii) Explain how the structure of ApoB-100 protein is maintained.

[3]

[Total: 12]

[Turn over

- 2 Fig. 2.1 shows the mode of action of the sodium-potassium pump found on the cell surface membranes of sweat glands.

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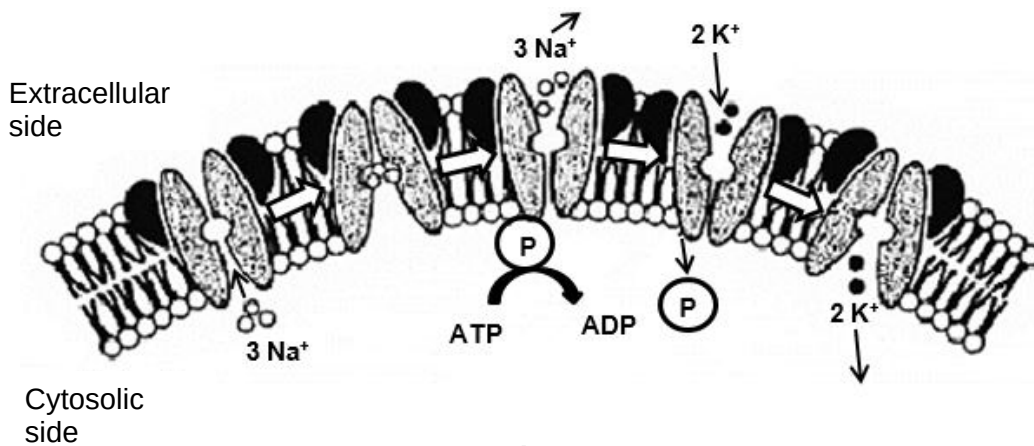


Fig. 2.1

- (a) Explain why the cell surface membrane is referred to as the 'fluid mosaic model'.

[2]

- (b) Explain why ions can only move across cell surface membranes via transmembrane proteins.

[2]

The sodium-potassium pump, also known as sodium-potassium ATPase, is able to catalyse the hydrolysis of ATP molecules to ADP and P_i . This reaction is coupled to the movement of ions across the membranes.

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Movement of ions is based on the initial binding of Na^+ ions to the pump. This stimulates phosphorylation of the pump which causes a change in its conformation, leading to ion movement.

(c) With reference to Fig. 2.1,

- (i)** identify the type of transport mechanism that enables sodium ions to leave the cell;

_____ [1]

- (ii)** explain how the pump is also able to catalyse the hydrolysis of ATP molecules leading to the movement of ions across the membranes.

_____ [3]

Ouabain is a cardiac glycoside that is able to affect the sodium-potassium pump. In an experiment to determine the effect of ouabain on the pump, sweat glands from rhesus monkeys were taken and treated to measure ATP hydrolysis.

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For each of the treatments, 5 sweat gland units were placed in a mixture of buffer and ATP.

Treatment 1 : Sweat glands of rhesus monkey with Na^+ and K^+ added.

Treatment 2 : Sweat glands of rhesus monkey with Na^+ , K^+ and ouabain added.

Treatment 3 : Sweat glands of rhesus monkey without additional Na^+ and K^+ .

Table 2.1 shows the data collected at the end of the 1 hour experiment.

Table 2.1

Treatment	Number of moles of ATP hydrolysed g^{-1}
1	0.58
2	0.37
3	0.38

(d) Explain the effects of ouabain on ATP hydrolysis.

 [2]

(e) Suggest a reason for the results of Treatment 3.

 [1]

[Total: 11]

[Turn over

- 3 Eggs and early embryos of amphibians are particularly suitable for biochemical studies of the cell cycle. These organisms typically have large eggs and fertilization is followed by multiple synchronous cell cycles.

As oocytes develop in the frog ovary, they become arrested in G_2 for 8 months, during which time they grow in size to a diameter of 1 mm. When stimulated by a male frog, the female's ovarian cells will secrete progesterone, which induces the oocytes to undergo meiosis I.

The oocytes will progress to Metaphase II, where they are arrested and await fertilization. Meiosis is completed only when the oocyte is fertilized by a sperm. Subsequently the first mitotic division of embryogenesis occurs. The entire process of oocyte maturation is shown in Fig. 3.1.

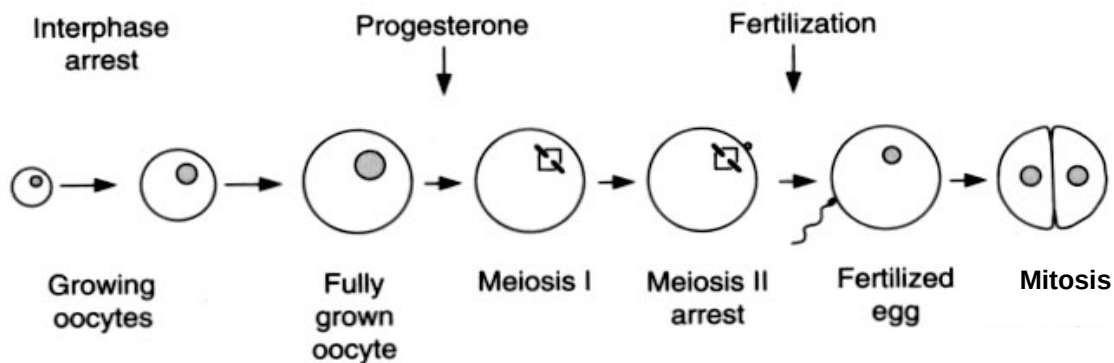


Fig. 3.1

- (a) G_2 is part of a stage that takes place during the cell cycle. Describe what happens during this stage.

[3]

(b) Distinguish the significance of meiosis from mitosis.

[3]

(c) Suggest why eggs of amphibians are suitable for studies of the cell cycle.

[2]

The Cama was created in a laboratory in Dubai, from breeding a male camel with a lama, with the purpose of higher wool production and easier temperament. Interestingly the cama, unlike most other types of animal hybrids, are always fertile.

(d) Suggest why most animal hybrids are sterile.

[3]

[Total: 11]

- 4 In bacteria, genes may be organised into operons. The *trp* operon consists of genes that encode five enzymes involved in the synthesis of tryptophan (*trp*). The *trp* operon and its regulatory gene are shown in Fig. 4.1.

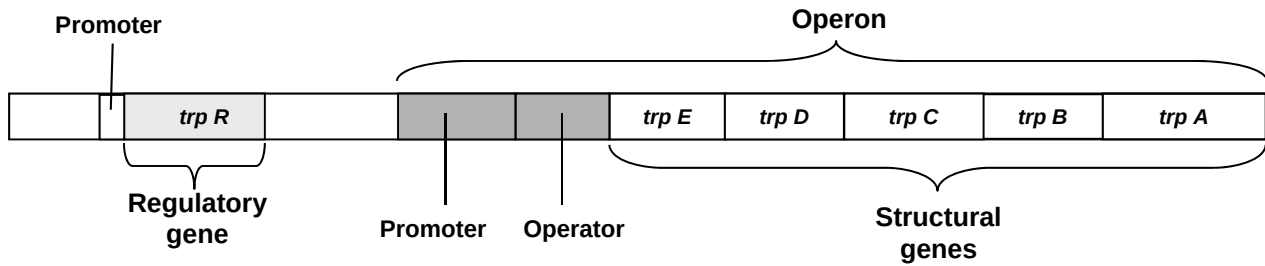


Fig. 4.1

- (a) With reference to the *trp* operon, explain what is meant by a *repressible operon*.

[3]

Different operons in the bacteria encode different types of enzymes.

- (b) State **one** difference between a repressible enzyme and an inducible enzyme.

[1]

Table 4.1 indicates the activity levels of the functional enzymes **E**, **D**, **C**, **B** and **A** in the presence and absence of tryptophan (Trp) in the culture medium of the different types of bacteria cells. The mutant strain is the result of a single base-pair substitution in a single component of the *trp* operon.

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Table 4.1

Enzymes	Activity level of enzymes / arbitrary units							
	Wild-type		Mutant strain 1		Mutant strain 2		Mutant strain 3	
	Trp absent	Trp present	Trp absent	Trp present	Trp absent	Trp present	Trp absent	Trp present
E	700	0	700	700	700	0	0	0
D	700	0	700	700	0	0	0	0
C	700	0	700	700	700	0	0	0
B	700	0	700	700	700	0	0	0
A	700	0	700	700	700	0	0	0

- (c) Identify and explain which mutant strain has a phenotype that is consistent with a loss-of-function mutation in the *trp R* gene.

[2]

- (d) Identify where the loss-of-function mutation occurred in the *trp* operon of mutant strain 3.

[1]

- (e) The eukaryote genome differs from the prokaryote genome in that some genes may be present in multiple copies as a result of gene amplification.

Describe the significance of gene amplification during development.

[2]

- (f) State **one** difference between the control of gene expression in eukaryotes and prokaryotes at the transcriptional level.

[1]

[Total: 10]

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Higher 2

Paper 2 Core Paper

Section A [Part 2]

Candidate Name:

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- 5 Fig. 5.1 shows a transmission electron micrograph of a bacterium.



(source: http://cronodon.com/BioTech/Bacteria_Growth.html)

Fig. 5.1

- (a) Describe and explain the significance of the process occurring in Fig. 5.1.

[3]

[Turn over

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This particular strain of bacteria carries an extra piece of genetic information encoded on an F plasmid.

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(b) Compare the F plasmid with the bacterial chromosome.

[3]

It has been shown that there is an increase in prevalence of antibiotic resistance among *E. coli* strains in Asia, and majority of the antibiotic resistances were due to the acquisition of antibiotic-resistance genes.

The effects of antibiotic **B** were studied on two strains of *E. coli*. The bacteria were cultured with antibiotic **B** and without antibiotic **B**. Fig. 5.2 shows the change in the number of bacteria cells.

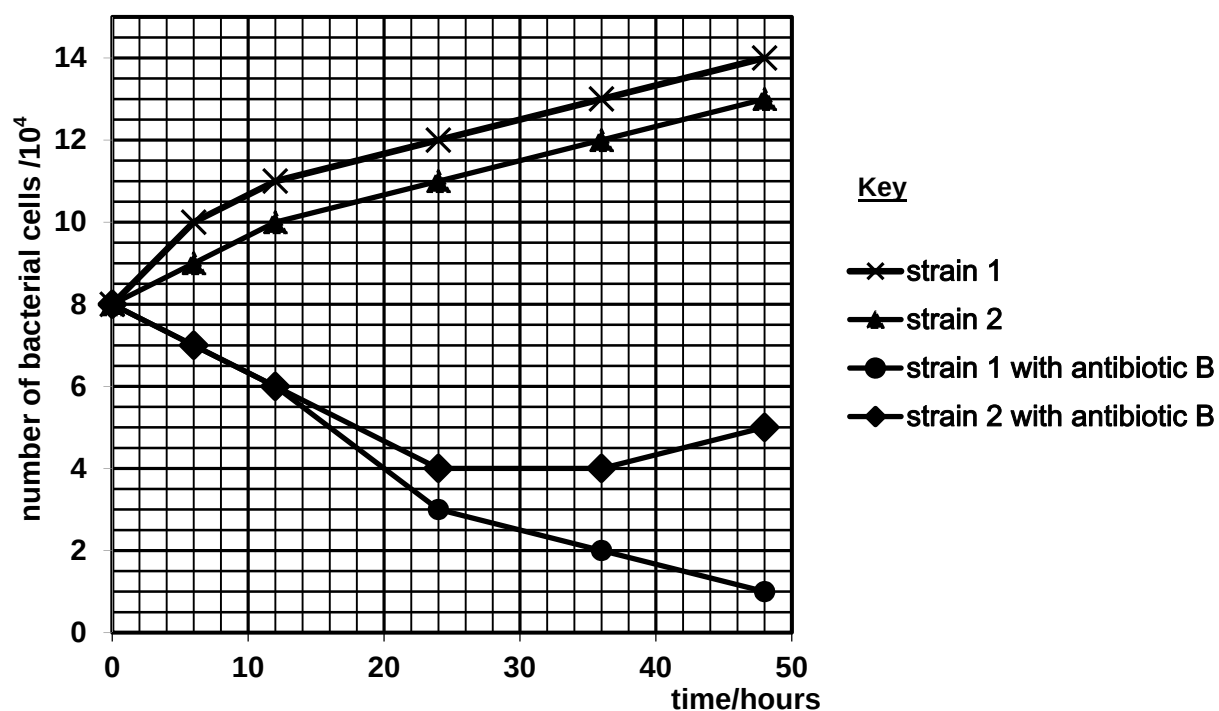


Fig. 5.2

(c) With reference to Fig. 5.2,

(i) describe the effect of antibiotic **B** on strain 1;

[2]

(ii) explain which strain contains the antibiotic resistance gene.

[2]

[Total: 10]

- 6 In the inheritance of feather colour in parakeets, there is an interaction between two autosomal gene loci, **A/a** and **B/b**. Parakeets carrying only the dominant allele, **A**, have yellow plumage while parakeets carrying only the dominant allele **B** have blue plumage. Parakeets with at least one dominant allele at both gene loci appear to have green plumage.

A parakeet with green plumage was crossed with a white parakeet. All the resulting progeny have green plumage. When pairs of the offspring were allowed to interbreed, they produced the following results in Table 6.1.

Table 6.1

Plumage colour	Number
Green	37
Yellow	11
Blue	12
White	4

- (a) Identify the type of gene interaction.

_____ [1]

- (b) State the possible genotypes of parakeets with blue plumage.

_____ [1]

(c) Draw a genetic diagram to explain the results of the cross between the offspring.

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[4]

[Turn over

- (d) Calculate the chi-squared (χ^2) value on the data to determine the probability of the results of the cross between the offspring departing significantly by chance from the expected ratio.

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

[2]

- (e) Explain if the results of the cross conform to the expected ratio.

[2]

[Total: 10]

[Turn over]

- 7 There are many species of the Anole lizards throughout the Caribbean and the surrounding mainland. Each species is found only on one island or a small group of islands, apart from *Anolis carolinensis* which is found in mainland Florida.

Fig. 7.1 shows the distribution of four species of anole lizards.

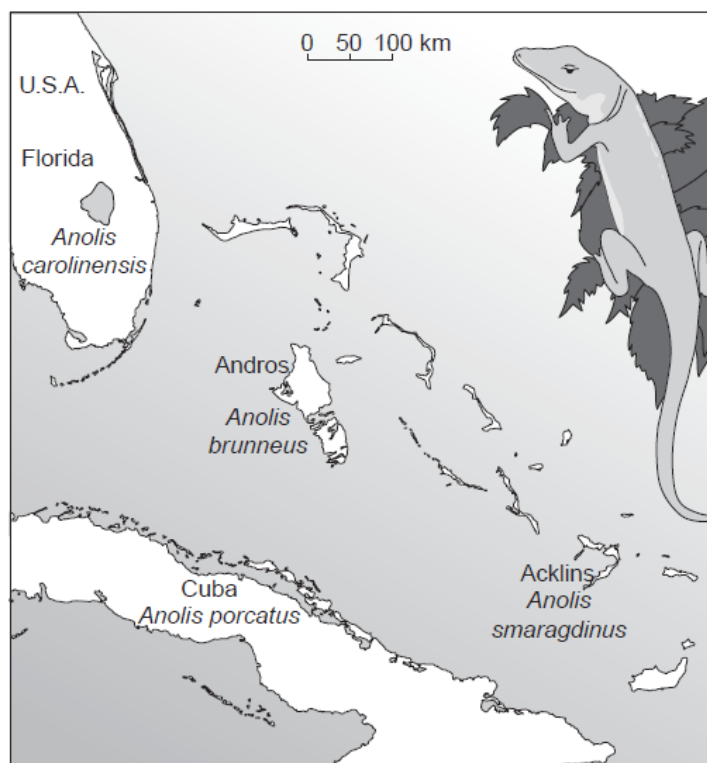


Fig. 7.1

An investigation using DNA analysis was carried out to establish the relationships between the four lizard species, using DNA analysis. The base sequences of a region of mitochondrial DNA from the four species were compared. The percentage differences in base sequences are shown in Table 7.1.

Table 7.1

	<i>A. brunneus</i>			
<i>A. brunneus</i>		<i>A. smaragdinus</i>		
<i>A. smaragdinus</i>	12.1		<i>A. carolinensis</i>	
<i>A. carolinensis</i>	16.7	15.0		<i>A. porcatatus</i>
<i>A. porcatatus</i>	11.3	8.9	13.2	

(a) With reference to Table 7.1,

(i) state the species to which *A. brunneus* appears to be least closely related;

_____ [1]

(ii) explain your answer in **(a) (i)**.

_____ [1]

(b) The researchers put forth the hypothesis that *A. brunneus*, *A. smaragdinus* and *A. carolinensis* all originated from *A. porcatus* in three separate events.

(i) Explain how the results in Table 7.1 support the hypothesis;

_____ [3]

(ii) Explain how a population of *A. porcatus* that became isolated on an island could evolve into each of the new species.

_____ [4]

[Total: 9]

- 8 Fig. 8.1 shows an electron micrograph of a cholinergic synapse.

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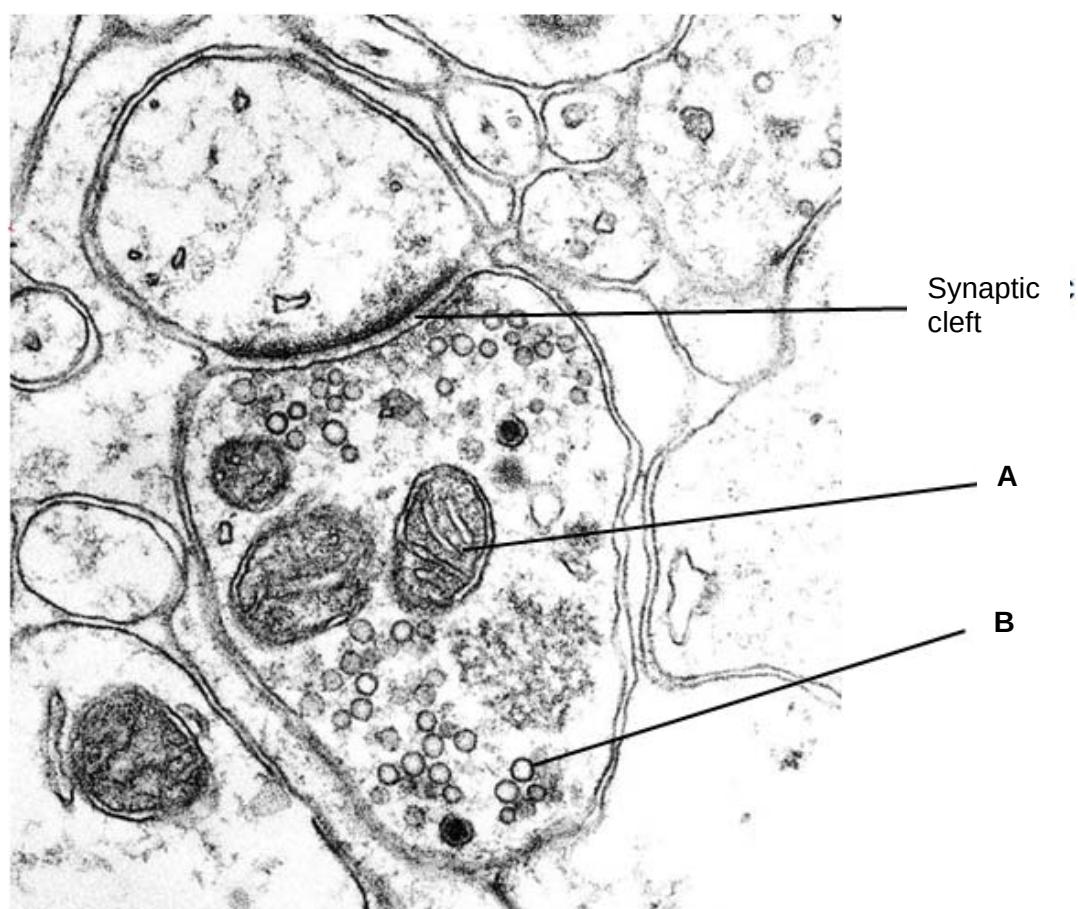


Fig. 8.1

- (a) Label the pre-synaptic membrane and post-synaptic membrane on Fig. 8.1. [1]

- (b) Identify what is present in structure B.

_____ [1]

- (c) Explain the role for structure A in the transmission of a nerve impulse across the synapse.

_____ [2]

[Turn over

Botulinum toxin is a neurotoxin that interferes with neural transmission, causing muscle paralysis. Botulinum can be used clinically by botox users to reduce wrinkles.

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The toxin is a protease which cleaves proteins called SNAREs. The primary role of the SNARE protein is to mediate fusion of vesicles to membranes.

(d) Explain how botulinum affects transmission of nerve impulse across the synapse.

[2]

(e) Suggest how botox is able to reduce wrinkles.

[1]

[Total: 7]

END OF SECTION A

[Turn over

Section B: Essay Questions

There are 2 questions in this section. Answer **only one** question.

Write your answer on the separate answer paper provided. At the end of the examination, fasten the separate answer paper securely.

You are advised to spend not more than 30 minutes on this section.

Your answer should be illustrated by large, clearly labelled diagrams wherever suitable.

- 9 (a) Describe how cellulose differs from starch. [5]
- (b) Describe how carbohydrates are formed in the Calvin cycle. [6]
- (c) Explain the effect of light and carbon dioxide as limiting factors on the rate of photosynthesis. [9]

[Total: 20]

- 10 (a) Describe how the structure of transfer RNA is adapted to its function. [5]
- (b) Describe how the information on an mRNA is used to synthesize a polypeptide chain. [8]
- (c) Discuss how the synthesis of a functional protein can be controlled at the translational and post-translational level. [7]

[Total: 20]

END OF SECTION B

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